Arab Academy of Science and Technology and Maritime Transport

COLLEGE OF ENGINEERING & TECHNOLOGY – Computer Engineering Dept.

Course: **Programming Fundamentals**

Course No.: **ECE1101**

TA: **Eng. Lobna Mohamed**

Sheet 1

If Conditional and Switch Statement

Write a program that:

1. Reads a number and tests if it is odd or even.

2. Reads a number from the user, checks if the number is **divisible by both 5 and 11**.

3. Takes a character from the user, checks if it is a vowel or consonant.

4. Displays the Grade Definition for the subjects that the user will enter its mark.(***use If***)
   Grading System:
   >=90: Excellent ..... >=79: V.Good ..... >=68-78: Good ..... >=50: Pass ..... Other: Fail

5. Takes two numbers and an operator (+, -, *, /) from the user, and based on the operator the user chose, perform the calculation. Display an error message if the operator is invalid. (***use Switch***)

6. Asks the user to enter a number between 1 and 7, implicitly, and print the corresponding day. (1→Monday, etc..). Display an error message if the number is out of range.

7. Displays a menu as follows:
   1-Circle
   2-Rectangle
   3-Square
   4-Triangle
   Then, asks the user to select a shape, and enter the required dimensions.
   Calculate and display the area of the selected shape.

8. Determine the values of the following conditional expressions.
   Note that, 0→ False, any other value→ T:
   a. ((size > 5) || (sum >= 150)) given size = 10, sum = 150
   b. ((cont) && (value > 'A')) given cont = 0, value = 'G'
   c. (!halt &&(sum < 100 ||test)) given halt = 0, sum = 150, test = 1

9. Convert the following If statements to Switch

```
if(x==1||x==2)
    cout<<"Low";
else if(x>=3 &&x<6)
    cout<<"Medium";
else if(x==6)
    cout<<"High";
else
    cout<<"Out of Range";
```

10. Trace the following switch statement, and show the output. (weather = 10).

```
switch (weather)
{
        case 30:
        case 40:
                cout << "Hot. \n";
                break;
        case 10:
        case 12:
                cout << "Cold.\n";
        case 0:
                cout << "Freezing.\n";
        default:
                cout << "Normal.\n";
}
```

11. Convert the following
    a. If statement to Switch statement

```
char grade;
cout << "Enter your grade (A, B, C, D, F): ";
cin >> grade;
if (grade == 'A' || grade == 'a')
    cout << "Excellent!";
else if (grade == 'B' || grade == 'b')
    cout << "Very Good";
else if (grade == 'C' || grade == 'c')
    cout << "Good";
else if (grade == 'D' || grade == 'd')
    cout << "Pass";
else if (grade == 'F' || grade == 'f')
    cout << "Fail";
else
    cout << "Invalid grade!";
```

b. Switch Statement to If statement

```
int month;
cout << "Enter month number (1-12): ";
cin >> month;
switch (month)
{
    case 1: case 3: case 5: case 7: case 8: case 10: case 12:
        cout << "This month has 31 days.";
        break;
    case 4: case 6: case 9: case 11:
        cout << "This month has 30 days.";
        break;
    case 2:
        cout << "This month has 28 or 29 days.";
        break;
    default:
        cout << "Invalid month number!";
}
```